

A provisional approach-conditions index from Mode S

Enhanced Surveillance replies on final approach at Helsinki-Vantaa: method, first results and runway comparison

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Abstract. Aircraft on final approach continuously correct for gusts and turbulence, and much of this activity can be observed from the ground. Air traffic control radars routinely request two Mode S Enhanced Surveillance registers. BDS 5,0 carries the aircraft's bank angle, and BDS 6,0 its indicated airspeed and vertical rate. This report describes how a single amateur 1090 MHz receiver near Helsinki-Vantaa airport and the open MODE-S Wind software turn these replies into a per-landing *approach-conditions index* on a 0–10 scale.

All replies received while an aircraft is established on final are collected for two altitude segments: 3000–1000 ft and 1000–200 ft MSL. Slow changes, such as deceleration and glideslope capture, are removed with a 15 s moving median. The remaining quick fluctuation of airspeed, vertical rate and bank angle is scored against calm and rough reference values. An aircraft-class factor accounts for the larger response of lighter aircraft.

The data set covers 473 landings over 49 h, from 6 October 12:30 to 8 October 13:40 UTC 2026. It includes a gusty westerly afternoon, two nights, a calm morning and a moderate easterly day, and five landing runways. The index follows the METAR surface wind ($\rho = 0.83$ per landing, $\rho = 0.88$ for hourly medians). Two landings less than 5 min apart give similar values ($\rho = 0.84$, median difference 0.5). The index has a pronounced diurnal cycle, from about 5 in the afternoon to 0.5–1 at night, even under a 25–45 kt low-level wind maximum. It therefore measures turbulence in a well-mixed boundary layer rather than the wind or the wind shear aloft.

Under matched conditions, turboprops scored 1.56 and regional jets 1.13 times the narrowbody level, and widebodies the same as narrowbodies. The class factors were corrected on 8 October; afterwards all classes lie within $\pm 6\%$ of the narrowbody level. After allowing for the surface wind, the five runways differed by less than half an index point. The clearest runway-specific effect is instrumental: the short finals of runways 04L and 04R lie on the far side of the airport from the receiver, and their lowest segment could be scored for only 57 % and 35 % of landings. A set of runway metrics is proposed for monitoring as the data grow. All results are preliminary.

1 Introduction

Gusts and turbulence close to the ground are among the most relevant weather factors for a landing aircraft, yet they are poorly observed. A METAR gives the 10-minute mean surface wind once every 30 or 60 minutes, and reports a gust value only when the variation is large. Between the surface and about 3000 ft there are usually no routine turbulence observations at all.

Every aircraft on final approach is itself a turbulence sensor. On a stabilised approach in calm air its airspeed, descent rate and bank angle are nearly constant. In gusty air they fluctuate within seconds, and the autopilot or the pilot corrects them. Mode S Enhanced Surveillance (EHS) replies, downlinked to air traffic control radars, contain enough information to derive wind and temperature [1]. Kopeć et al. [2] showed that atmospheric turbulence can also be retrieved from regular commercial aircraft using Mode S and ADS-B data.

The MODE-S Wind project runs on a single amateur receiver and already derives wind profiles on final approach at Helsinki-Vantaa (EFHK). Since 6 October 2026 it also stores a short roughness summary of every established final approach and converts it into one number, the *approach-conditions index*.

Working version 1 of this report described the method and the first 31 hours. Version 2 extends the data to 49 hours, including a change from westerly to easterly wind and four further landing runways. It also documents the tuning steps of

7 and 8 October and adds a first comparison between runways.

Sections 2–3 describe the system and method, Sections 4–6 the general results and aircraft classes, and Section 7 the runways. The report is a starting point for a fuller study once months of data are available.

2 Receiving system and software

Receiver and antenna. The data come from a Jetvision Radarcape 1090 MHz Mode S/ADS-B receiver in Tuusula, roughly 10 km north of EFHK. The antenna is a ground-plane (GP) antenna for 1090 MHz on a mast about 7 m above ground. It is fed through 15 m of Ecoflex-10 coaxial cable. The cable attenuates about 15 dB per 100 m at 1090 MHz [5], so the 15 m run loses roughly 2.3 dB.

The receiver is passive. It hears the replies that aircraft transmit when the Mode S radars of the Finnish air navigation service provider interrogate them, as well as their ADS-B extended squitters. The Radarcape provides the raw replies as a Beast binary stream over TCP. It also provides a JSON aircraft list that includes multilateration (MLAT) positions, which remain available when GNSS-based ADS-B positions are degraded by jamming.

Processing. MODE-S Wind is written in Python and runs on a Raspberry Pi 4 (4 GB) on the same local network (Fig. 1). Replies are decoded with pyModeS [3]. BDS 4,4 meteorological reports are used directly. Most wind observations,

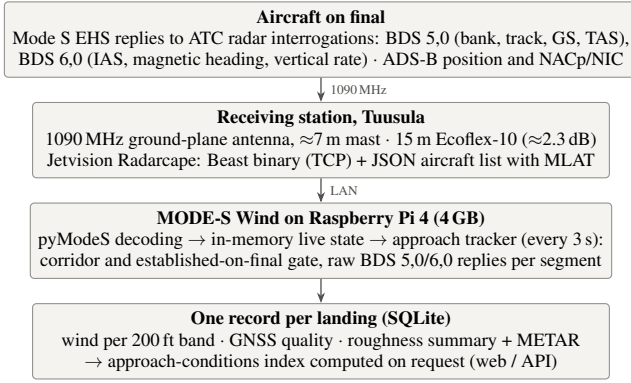


Figure 1. Data flow from the aircraft to the per-landing approach-conditions index.

however, are computed as the difference between two vectors. The ground vector is the true track and groundspeed from BDS 5,0. The air vector combines the magnetic heading from BDS 6,0, corrected with the World Magnetic Model declination, with the true airspeed.

An approach tracker detects aircraft established in the approach corridor of each EFHK runway. For every completed landing, one row is written to an SQLite database. It holds the wind in 200 ft altitude bands, the GNSS quality on final and, since 6 October 2026, the roughness summary described below. A web interface shows the live approaches, the per-landing history and the index. The index is computed when the data are requested and is never stored, so re-tuning its constants re-scores all past landings.

3 Method

3.1 Established on final and altitude segments

Only replies received while the aircraft is *established on final* are used. The aircraft must be in a runway’s approach corridor and before the threshold, not in a go-around. It must be within 1 NM of the extended centreline, with its track within 20° of the runway heading. The replies are collected separately for two segments. The *hi* segment covers 3000–1000 ft MSL, where the autopilot is usually engaged. The *lo* segment covers 1000–200 ft MSL and is often hand-flown.

Replies that the wind calculation would reject because of bank angle or turn rate are deliberately kept, since they show the gust response. Replies whose track or groundspeed disagrees with the aircraft’s ADS-B state, which indicates a mis-identified Comm-B register, are discarded. This was rare (median 0 per landing). The *hi* segment had a median of 105 BDS 6,0 and 72 BDS 5,0 replies over 136 s. The *lo* segment had 20 and 16 replies over 49 s.

3.2 High-pass fluctuation

The normal course of an approach also changes these quantities, through deceleration, flap-speed steps, glideslope capture and the end of the intercept turn. These changes take tens of seconds, whereas gust response takes a few seconds. For each quantity x with replies x_i at times t_i , the slow part is removed with a moving median and the root-mean-square (RMS) of

Table 1. Quantities used, their register and resolution, and the index constants (calm reference a , rough reference b , weight w).

Quantity	Register	Resolution	a	b	w
Indicated airspeed	BDS 6,0	1 kt	0.6 kt	3.0 kt	0.45
Vertical rate ¹	BDS 6,0	32 ft min ⁻¹	30	160	0.30
Bank (roll) angle	BDS 5,0	0.18°	0.4°	2.5°	0.25
Crab angle ²	6,0 + track	0.18°	—	—	0

¹Inertial vertical velocity, or the barometric altitude rate if it is missing.

²Magnetic heading (BDS 6,0) + declination – current true track. Stored for information only.

the remainder is taken:

$$r_i = x_i - \text{median}\{x_j : |t_j - t_i| \leq \tau\}, \quad \tau = 7.5 \text{ s}, \quad (1)$$

$$\tilde{x} = \left(\frac{1}{N} \sum_{i=1}^N r_i^2 \right)^{1/2}. \quad (2)$$

The median makes the reference robust to single bad replies. A value is computed only from at least 8 replies. This gives $\tilde{x}_{\text{IAS}}$ in kt, $\tilde{x}_{\text{VR}}$ in ft min⁻¹ and $\tilde{x}_{\text{bank}}$ in degrees per segment. The plain standard deviations are stored as well. The first 12 landings of 6 October were logged before the high-pass values existed. For them the plain values are used, with their own references.

3.3 Index

Each component c is scored linearly between its calm and rough reference (Table 1) and clipped:

$$S_c = \min\left(10, \max\left(0, 10 \frac{\tilde{x}_c - a_c}{b_c - a_c}\right)\right). \quad (3)$$

The segment score is the weighted mean $S_{\text{seg}} = \sum_c w_c S_c / \sum_c w_c$ over the available components. The landing score R (“raw”) is the mean of the scored segments. The index is

$$I = \min(10, R/f_{\text{class}}), \quad (4)$$

where f_{class} is taken from the ICAO type code: 1.5 for turbo-props, 1.25 for business jets, 1.15 for regional jets and 1.0 for narrowbodies and widebodies. The levels are Smooth ($I < 2$), Light (2–4), Choppy (4–6), Rough (6–8) and Very rough (≥ 8). These terms deliberately differ from the ICAO turbulence categories, because I is not a turbulence report.

3.4 Tuning history

All constants were set by hand and adjusted twice against the data presented here (Table 2). The results in Sections 5–6 therefore describe the fit and are not an independent validation (Section 8). Because the index is computed on request, every figure in this report uses the constants in force on 8 October.

3.5 Reference data

The roughness record stores the METAR wind valid at landing time. Hourly EFHK METARs were also retrieved from the Iowa Environmental Mesonet archive [6], because the gusts on 6 October were reported only in TEMPO and BECMG trend groups. In addition, each landing record holds the aircraft’s own computed wind in 200 ft bands. These were vector-averaged over 200–1000 ft and 1200–3000 ft. A smooth reference curve of index against METAR wind was fitted to all

Table 2. Tuning history of the index.

Date	Change and basis
6 Oct	First scale from one gusty afternoon (METAR 280/18–21G32 kt), using the plain statistics and the crab-angle variation. High-pass values added from about 16 UTC.
7 Oct	Re-tuned on 236 landings (METAR 6–20 kt). High-pass references as in Table 1, weights 0.45/0.30/0.25. Crab-angle variation reported only (no relation to the wind). The scale was widened because it had been compressed (median 2.4 at 13–16 kt).
8 Oct	Class factors measured on 356 landings (Section 6): turboprop 1.35→1.5, widebody 0.85→1.0. Regional jet (1.15) and business jet (1.25) unchanged.

landings (LOWESS, span 0.5). It gives 0.6 at 2 kt, 1.0 at 5 kt, 2.4 at 10 kt, 3.5 at 12 kt and 4.3 at 15 kt, and is used to compare runways at equal wind.

4 Data

From 6 October 2026 12:30 UTC to 8 October 13:40 UTC the roughness log contains 481 records. Four helicopter records were excluded and four landings had too few replies to be scored, leaving 473 landings. Both segments were scored for 437 landings, only the hi segment for 33 and only the lo segment for 3.

By class there were 229 narrowbodies, 96 turboprops (89 ATR 72), 79 regional jets (65 E190), 43 widebodies, 8 business jets and 18 aircraft of unknown type, which are scored as narrowbodies. By runway there were 326 landings on 22L, 19 on 22R, 68 on 15, 40 on 04L and 20 on 04R.

The weather passed through four phases (Fig. 2):

- **6 Oct, afternoon:** westerly post-frontal flow, CAVOK, METAR 27021G32KT at 10:50 UTC and TEMPO gusts of 26–31 kt until 16 UTC. Runway 22L.
- **7 Oct:** westerly 12–15 kt by day and 4–7 kt at night. Runway 22L, with 22R in parallel at times.
- **Night of 7–8 Oct:** calm (VRB01–02KT, near-freezing temperature). Wind at 1200–3000 ft decreased from about 20 to 7 kt. Runways 22L, then 15.
- **8 Oct:** the wind turned easterly (070–110°) and freshened to 10 kt, with the first METAR gust in the main report at 13:50 (10011G21KT). Runway 15 until 09 UTC, then briefly 22L, and from 11 UTC 04L and 04R in parallel.

5 Results

5.1 Diurnal cycle

The hourly median index (Fig. 2) was 4.9 over 12–16 UTC on 6 October. It fell to 1.1 during the following night (21–05 UTC) and rose to 4.3 over 08–13 UTC on 7 October. During the calm night of 7–8 October it fell to 0.5 (19–05 UTC). On 8 October it rose to 1.6 in the light easterly morning on runway 15 (06–10 UTC) and to 2.3 on runways 04L/04R in the afternoon (11–14 UTC, METAR 10 kt).

The highest single value was 8.2, for an A319 at 13:13 UTC on 6 October (METAR 27018KT, TEMPO gusts to 28 kt). Over the whole set, 209 landings were Smooth, 149 Light, 109 Choppy, 5 Rough and 1 Very rough.

Table 3. Index by METAR mean surface wind (median and interquartile range).

METAR wind	Landings	Median <i>I</i>	IQR
1–4 kt	92	0.7	0.4–1.0
5–8 kt	117	1.2	0.6–2.2
9–12 kt	144	2.6	1.9–3.8
13–16 kt	95	4.2	3.6–4.7
17–21 kt	25	4.5	4.1–5.2

The first night shows that the index does not simply follow the wind. The surface wind was only 7 kt (median), but the landing aircraft measured 26 kt at 200–1000 ft and 35 kt at 1200–3000 ft. This is a typical nocturnal low-level wind maximum above a stable surface layer [4]. Despite the strong wind and the strong shear below 1000 ft, the approaches were smooth.

Across the whole set, the median index was 3.3 by day (sun more than 5° above the horizon, 253 landings) and 0.9 at night (sun more than 5° below, 182 landings). The ratio of surface wind to 200–1000 ft wind is a simple indicator of how well the lowest layer is mixed. It was 0.64 by day and 0.28 at night (medians).

5.2 Relation to the wind

Across all landings the index correlates with the METAR mean wind with Spearman $\rho = 0.83$ (Fig. 3, Table 3). For hourly medians with at least three landings the correlation is 0.88 (46 hours). Within each regime the relation holds as well, with $\rho = 0.75$ by day and 0.71 at night. The index also correlates with the surface-to-200–1000 ft wind ratio ($\rho = 0.72$).

The wind measured by the aircraft aloft is a poor predictor. Over the first 31 h the index correlated slightly *negatively* with it ($\rho = -0.21$ at 200–1000 ft and -0.31 at 1200–3000 ft), because of the windy but smooth first night. With the calm second night included, the correlation is weakly positive (0.19 and 0.21). The vector wind difference between the surface and 200–1000 ft correlates negatively with the index ($\rho = -0.44$): strong shear near the ground occurred mainly in stable, smooth conditions.

5.3 Which components carry the signal

All three high-pass components follow the surface wind in both segments (Fig. 4). Airspeed is strongest ($\rho = 0.77$ hi, 0.82 lo), followed by bank angle (0.50, 0.66) and vertical rate (0.39, 0.64). In the hi segment the plain statistics are much weaker (0.04–0.20), because the slow changes of the intercept and of glideslope capture dominate them. The moving median is what makes the signal visible there. In the short lo segment the plain and high-pass values behave similarly.

The crab-angle variation now shows a weak positive relation (0.35 hi, 0.30 lo), whereas in the first 31 h it showed none. It remains unscored for now.

5.4 Repeatability

For consecutive landings less than 5 min apart (308 pairs), the index values correlate with $\rho = 0.84$, with a median absolute difference of 0.5 index points. The hi and lo segment scores of the same landing correlate with $\rho = 0.71$ and are on average

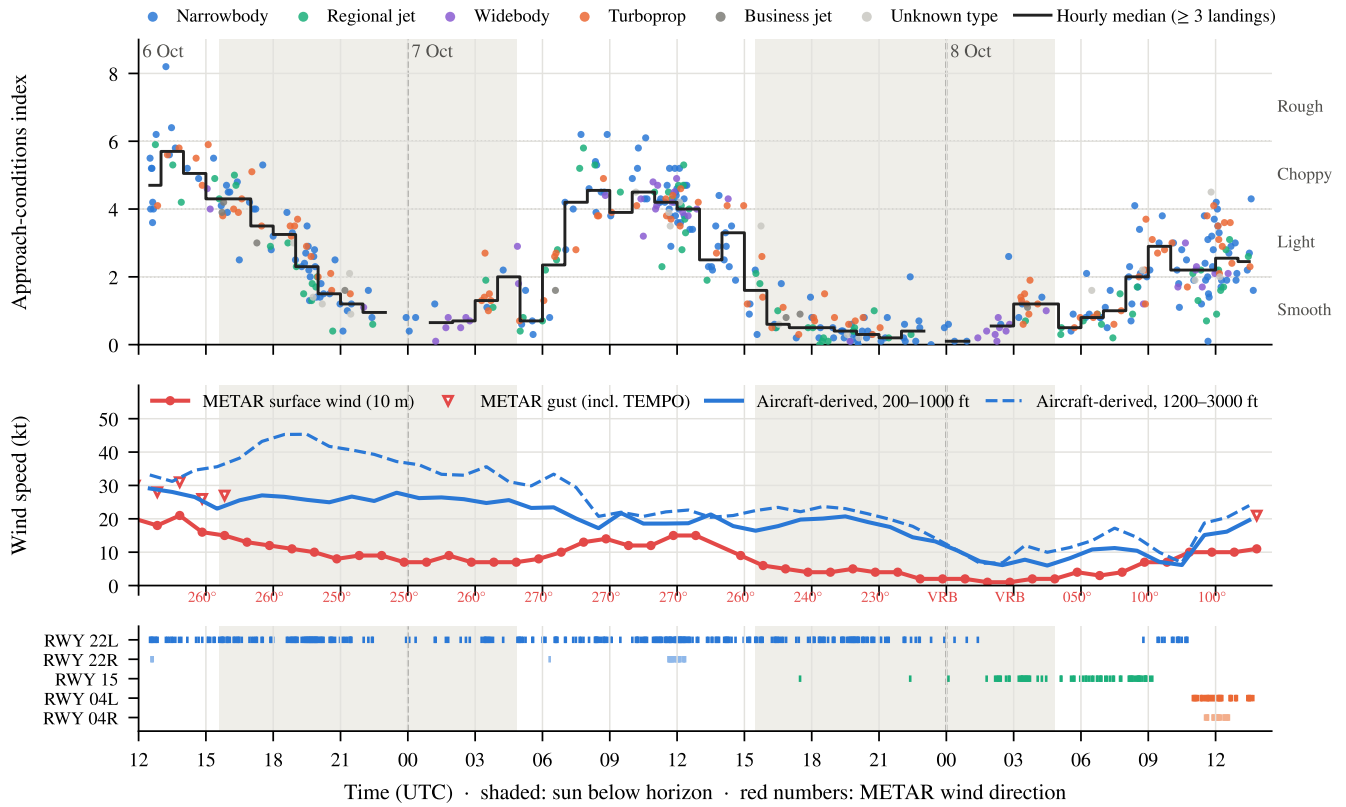


Figure 2. Top: approach-conditions index of every landing, coloured by aircraft class, with the hourly median (black). Middle: METAR surface wind and gust, and the wind measured by the landing aircraft themselves, as hourly medians of the vector-mean wind at 200–1000 ft and 1200–3000 ft. The METAR wind direction is printed every three hours. Bottom: landing runway of each landing. Grey shading marks the time the sun was below the horizon at EFHK.

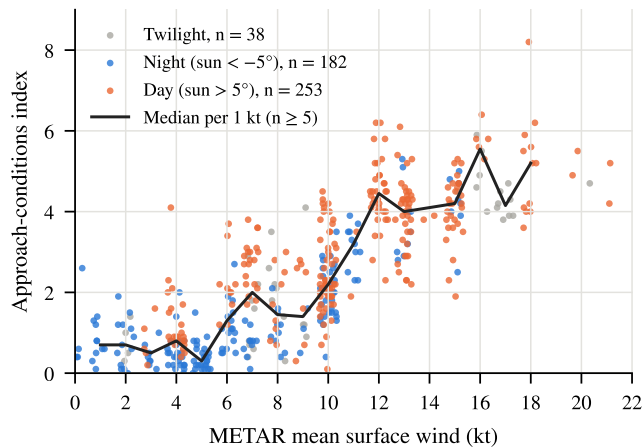


Figure 3. Index against the METAR mean surface wind, coloured by sun elevation. A small horizontal jitter separates the 1 kt METAR values.

about equal (median difference -0.1). A single landing is therefore a noisy but useful sample, and a median over a few landings is a robust description of the current conditions on final.

6 Aircraft classes and types

Light aircraft are moved more by the same gust: the response grows with airspeed and lift-curve slope and decreases with wing loading. The class factor is meant to remove this effect.

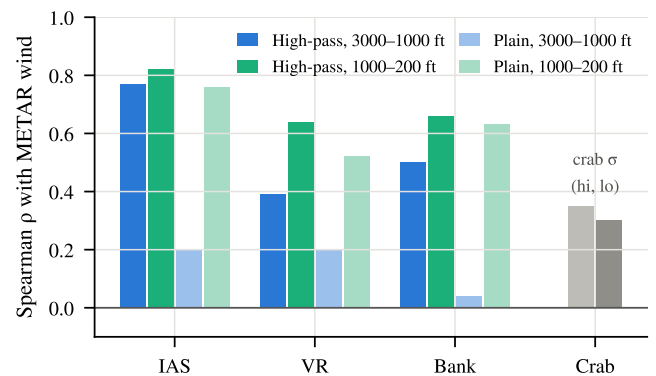


Figure 4. Spearman correlation of each component with the METAR mean wind, for the two segments. “High-pass” is the RMS about the 15 s moving median used by the index. “Plain” is the standard deviation (for IAS, about a linear trend) or, for bank angle, the RMS.

Because the classes are not spread evenly over the day, comparing their plain averages would mostly compare day and night. Instead, the raw score of each landing was divided by the median raw score of all other landings within ± 45 min, which share the same weather. Hours with a local median below 1.5 were left out, because there the ratio is dominated by noise. This left 328 landings (Fig. 5, Table 4).

The first 31 h (230 matched landings) gave turboprop 1.55, regional jet 1.20 and widebody 1.01. The factors were changed accordingly on 8 October: turboprops from 1.35 to 1.5, and widebodies from 0.85 to 1.0, because widebodies

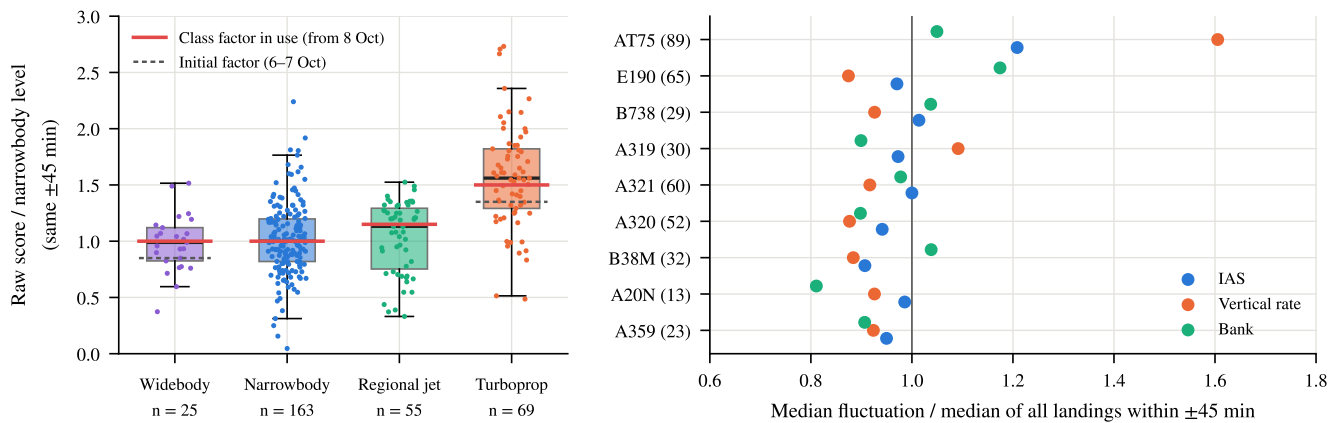


Figure 5. Left: raw score R (before the class factor) of each landing, divided by the median raw score of all other landings within ± 45 min and normalised to the narrowbody median. Only landings with at least four others in the window and a local median $R \geq 1.5$ are shown. Red bars are the class factors in use since 8 October and dashed bars the initial ones. Right: the same matched comparison for each component and aircraft type (median ratio; number of landings in brackets).

Table 4. Matched raw-score level of each class relative to narrowbodies (median, bootstrap 95 % interval), and the class factors.

Class	n	matched	observed	initial	in use
Turboprop	96	69	1.56 (1.43–1.70)	1.35	1.5
Regional jet	79	55	1.13 (0.96–1.26)	1.15	1.15
Narrowbody	229	163	1.00	1.00	1.00
Widebody	43	25	0.99 (0.86–1.10)	0.85	1.0
Business jet	8	5	1.01 (0.72–1.21)	1.25	1.25

had been scored about 15 % too high. The result was the same with ± 30 or ± 60 min windows and with local-median thresholds of 0.5 or 2.5. The extended data confirm the change. With the factors now in use, the matched index of each class lies within ± 6 % of the narrowbody level: turboprop 1.06, regional jet 0.96 and widebody 0.97. Only business jets (0.81) stand out, and they are based on just 4 matched landings, so their factor has been left unchanged.

The right panel of Fig. 5 shows that the types differ in *how* they respond, not only in *how much*. The ATR 72 shows mainly larger vertical-rate (1.61) and airspeed (1.21) fluctuation, consistent with its lower wing loading and approach speed. The E190 shows larger bank-angle activity (1.17) but less vertical-rate fluctuation (0.87). The A320neo and the A350-900 show the least bank-angle activity (0.81 and 0.91). A single scalar factor per class can therefore only partly normalise the index. Per-type normalisation is the natural next step once each type has enough landings.

7 Runway comparison

7.1 Why runways might differ

Each EFHK landing direction has its own final approach over different terrain, at a different distance and direction from the receiver. Physical runway 04R/22L is used in both directions, as is 04L/22R; runway 15 is the cross runway. The approach to 22L/22R comes from the north-east, to 15 from the north-west and to 04L/04R from the south-west.

Runway-specific behaviour could arise in several ways:

- different upstream terrain, forest, built-up areas or the

airport itself in a given wind direction;

- different crosswind and headwind for the same wind;
- differences in the approach procedures;
- different reception geometry, which changes how many replies are collected.

The runway in use, however, is chosen by the wind and the time of day. In this data set each runway was used in essentially one weather situation (Table 5). Runway and weather cannot yet be separated, so the comparison below is a first look and a template for monitoring.

7.2 Level at equal wind

At equal surface wind the runways differ little (Fig. 6a,b). The median wind-adjusted index lies between -0.21 (22L) and $+0.40$ (22R), less than the 0.5-point landing-to-landing scatter. Runways 15 and 22R lie slightly above the reference. For 15 this comes from the early morning of 8 October (03–04 UTC), when the index reached 1.2 at a surface wind of only 2 kt. At that time the aircraft measured 8–12 kt from the south-west aloft above an almost calm surface (VRB02KT). The daytime easterly approaches to 04L/04R (2.3 and 2.15 at 10 kt) lay on the reference curve. The westerly daytime approaches to 22L at similar winds (7–12 kt, 63 landings) were about 0.7 points above it. Whether this reflects the wind direction, the air mass (a cold post-frontal flow on 6–7 October) or the runway cannot be told from one day of each.

7.3 Parallel runways 04L and 04R

On 8 October between 11 and 14 UTC both 04 runways were used at the same time, which gives a matched comparison. The index was 2.3 on 04L and 2.15 on 04R, and the hi segment 3.0 and 2.65. Neither difference is significant (Mann–Whitney $p = 0.31$ and 0.28). The two parallel final approaches, flown at the same time, give the same picture, as expected.

7.4 Vertical structure

The lo and hi segments are on average equal, but not on every runway (Fig. 6c). On 22L in the mixed westerly flow the lo segment scored slightly higher than the hi segment (median difference $+0.2$). On runway 15, used mainly at night and in the calm morning, the lo segment scored much lower (0.55

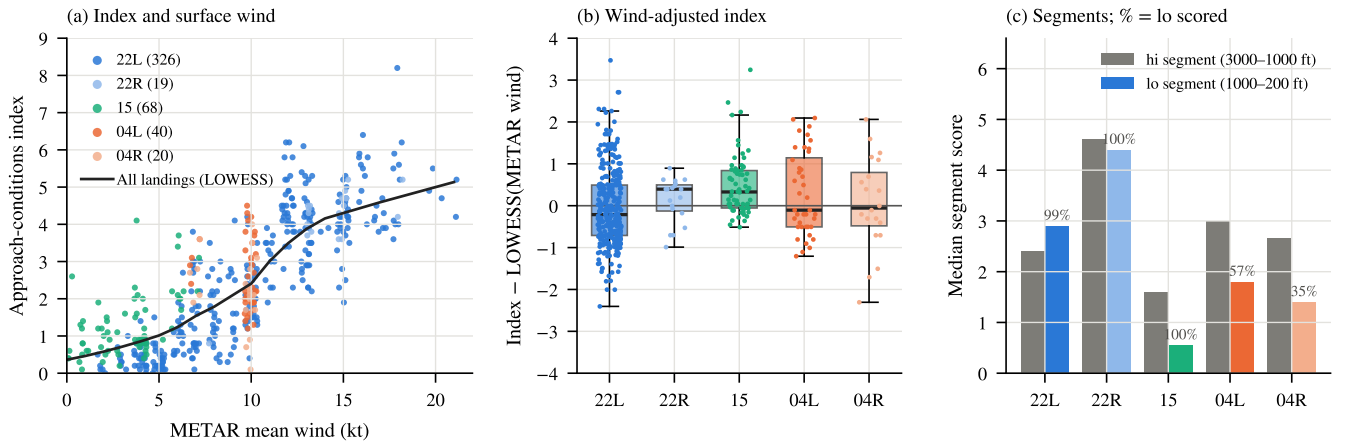


Figure 6. Runway comparison. (a) Index against METAR wind, coloured by landing runway (number of landings in brackets), with the LOWESS reference curve of all landings. (b) Wind-adjusted index: the difference between each landing and the reference curve. (c) Median hi and lo segment scores per runway; the percentage is the share of landings whose lo segment could be scored.

Table 5. Runway summary, 6–8 October 2026. METAR wind, headwind (HW) and crosswind (XW, + from the right) are medians at landing time. Wind-adj. is the median difference from the LOWESS reference curve, and its IQR measures the spread. Reply rates are BDS 6,0 replies per second while established.

RWY	<i>n</i>	Used	METAR	HW/XW (kt)	Day	Index (IQR)	Wind-adj. (IQR)	hi	lo	lo scored	Rate hi/lo	Δ
22L	326	6–8 Oct	260/10	8.8 / +5.5	46 %	2.55 (0.8–4.1)	−0.21 (1.20)	2.4	2.9	99 %	0.77 / 0.40	0.5
22R	19	6–7 Oct, day	270/15	10.9 / +10.2	100 %	4.4 (3.9–4.7)	+0.40 (0.62)	4.6	4.4	100 %	0.66 / 0.37	0.4
15	68	7–8 Oct, night–morning	070/4	0.6 / −3.9	62 %	0.95 (0.7–1.7)	+0.33 (0.89)	1.6	0.55	100 %	1.02 / 0.59	0.45
04L	40	8 Oct, 11–14 UTC	100/10	4.5 / +8.0	100 %	2.3 (1.9–3.2)	−0.11 (1.65)	3.0	1.8	57 %	0.61 / 0.28	0.6
04R	20	8 Oct, 11–13 UTC	100/10	6.0 / +8.0	100 %	2.15 (1.9–2.8)	−0.06 (1.28)	2.65	1.4	35 %	0.55 / 0.25	0.8

|Δ|: median absolute index difference between consecutive landings on the same runway less than 5 min apart.

against 1.6). In those hours the wind turned from an almost calm, easterly surface wind to 10–17 kt from the south-west to south above. The aircraft thus passed a sheared, slightly turbulent layer aloft above a calm and stable surface layer. This is the kind of runway- and time-specific structure that the two segments can reveal as data accumulate.

7.5 An instrumental effect: reception of the lowest segment

The clearest difference between runways is not meteorological. The short finals of 04L and 04R lie south-west of the airport, on the far side from the receiver. There, the replies below 1000 ft arrived at only 0.25–0.28 per second, against 0.40 on 22L and 0.59 on 15. As a result, only 57 % (04L) and 35 % (04R) of the landings reached the 8 replies needed to score the lo segment. The hi segment was received well on all runways (0.55–1.0 replies per second). GNSS degradation on final was similar for all runways (7–11 % of samples) and does not explain the difference.

This affects the index in two ways. First, a landing without a lo segment is scored on the hi segment alone. On 04L these landings had a median index of 3.1, against 2.1 for landings with both segments. Second, sparse sampling biases the high-pass RMS slightly low. In a simulation with a gust-like signal (3 s correlation time), the RMS at 0.25 replies per second was about 10 % lower than at 1 reply per second. Within runway 22L, the lo-segment airspeed fluctuation showed no dependence on the reply rate ($\rho = 0.05$), so the second effect is small in practice. Comparisons between runways should nevertheless use the hi and lo segments separately and note the lo coverage.

7.6 Spread

The spread of the wind-adjusted index (its IQR) was largest on 04L (1.65) and smallest on 22R (0.62) and 15 (0.89). The consecutive-landing difference was 0.4–0.8 on all runways. The larger spread on 04L follows mostly from the mix of one- and two-segment landings described above. No runway yet shows a spread that is clearly meteorological.

7.7 What to monitor

With weeks to months of data, each runway will be used in several wind directions, speeds and times of day. The following metrics could then reveal runway-specific behaviour, for example a recognisable signature for runway 15 or 04L:

1. the wind-adjusted level per runway, separately for day and night and per wind-direction sector, against a reference model of surface wind and stability;
2. the spread per runway (IQR of the wind-adjusted index and the consecutive-landing difference), as an indicator of intermittent, locally generated turbulence;
3. the lo–hi segment difference per runway and wind direction, as an indicator of near-ground (terrain or obstacle) turbulence or of a stable surface layer;
4. the component signature per runway, for example more bank activity with crosswind from a particular side;
5. matched comparisons whenever two runways are used at the same time (04L/04R, 22L/22R, or 15 with another runway);
6. the lo-segment coverage and reply rate per runway, as a data-quality flag;
7. the go-around rate and the 200 ft wind-band shear per

runway, alongside the index.

8 Discussion

What the index measures. The index behaves like a measure of mechanically and convectively mixed boundary-layer turbulence. It rises with the surface wind, is high in the sunlit, well-mixed afternoon, and drops at night even under a 35–45 kt low-level wind maximum and strong shear. This is physically plausible: a stable stratification suppresses turbulence even when the shear is strong [4]. The index is not a wind-shear detector. Shear on final is better described by the 200 ft wind bands that MODE-S Wind already records.

Air versus aircraft. The index measures how much the aircraft was moved and corrected, not the air alone. Autopilot versus hand flying, approach speed, flap setting and autothrust all contribute. The two segments partly separate the autopilot from the hand-flown part, and type-specific response is substantial (Section 6). A physically based quantity, such as the derived gust velocity or the eddy dissipation rate (EDR) estimated from the same replies [2], would remove more of the aircraft from the measurement.

Sampling and resolution. Replies arrive irregularly, only while a radar interrogates the aircraft. As Section 7 shows, the rate depends on where the aircraft is relative to the receiver. The register resolution also sets a noise floor. For uniform quantisation the RMS error is the resolution divided by $\sqrt{12}$: about 0.3 kt for airspeed, 9 ft min⁻¹ for vertical rate and 0.05° for bank angle. This floor is well below the calm references but not negligible for airspeed in calm conditions, which partly explains why the index rarely reaches zero even at 1–2 kt surface wind.

Limitations. The data set covers 49 hours and essentially four weather situations, each tied to one runway configuration. The index constants were tuned on most of the same landings, so the correlations describe the fit rather than validate it. METAR is a weak reference for turbulence: it gives a 10-minute mean surface wind once per hour and reports gusts only when they are large. The class and type comparisons rest on tens of matched landings, and the runway comparison on a few hours per runway.

9 Conclusions

1. Mode S EHS replies received by a single amateur station are enough to characterise the roughness of practically every landing at EFHK, with about 100 replies per landing between 3000 and 1000 ft.
2. Removing slow changes with a 15 s moving median is essential above 1000 ft. There, the high-pass fluctuation of airspeed, vertical rate and bank angle follows the surface wind, whereas the plain statistics largely do not.
3. The index correlates with the METAR wind at $\rho = 0.83$ per landing and $\rho = 0.88$ per hour, and is repeatable between consecutive landings ($\rho = 0.84$, median difference 0.5).
4. The index has a clear diurnal cycle: Choppy in the after-

noon, Smooth at night even under a 45 kt low-level wind maximum. It reflects boundary-layer mixing, not wind speed or shear aloft.

5. Turboprops respond about 1.56 times as much as narrowbodies, regional jets about 1.13 times, and widebodies equally. With the factors corrected on 8 October (turboprop 1.5, widebody 1.0), all classes agree within $\pm 6\%$.
6. At equal surface wind the five runways differed by less than 0.5 index points so far. The parallel runways 04L and 04R gave the same result. The clearest runway effect is the reduced reception of the lowest segment on 04L/04R, which must be taken into account in runway comparisons.

10 Outlook

- Collect several months of data covering all runway configurations in varied winds, summer convection, winter inversions and real gust events. Then repeat the runway metrics of Section 7.
- Validate against independent references: the Finnish Meteorological Institute's 10-minute wind and gust observations at EFHK, radiosonde stability (Jokioinen, Tallinn) and the go-around rate.
- Add a reference model of the index against surface wind and a stability indicator, so that the wind-adjusted level can be computed per runway and shown live.
- For a limited period, store the raw reply series of each landing (about 1–2 kB per landing). Alternative methods could then be tested on the same landings: derived gust velocity, spectral analysis at 0.05–0.3 Hz and EDR [2].
- Replace the hand-set references and class factors with data-driven normalisation per type and component, and report an uncertainty and the segment coverage with each index value.
- Report separate quantities for turbulence (this index), wind shear (from the 200 ft wind bands) and crosswind.

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